

B. Tech. (6th Sem) Computer Science & Engineering with Specialization in AI & Machine Learning
BCSE-803L (Reinforcement Learning Lab)

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Continuous evaluation 60
End semester exam 40
Total marks 100
Credits 2.0

Course Objectives:

1. Understand core concepts of reinforcement learning, including agents, environments, states, actions, and rewards.
2. Implement fundamental RL algorithms such as Q-learning, SARSA, Value Iteration, and Policy Iteration.
3. Develop and test RL agents in standard environments using Python and libraries like OpenAI Gym.
4. Analyze agent performance through reward tracking, learning curves, and parameter tuning.
5. Apply RL techniques to solve real-world inspired problems and build simple AI projects.

List of Practical

1. Installation of Code Standards and Libraries used in Reinforcement Learning (Python/Keras/Tensorflow).
2. Write a program to solve Multi Armed Bandit Problem.
3. Write a program to implement Dynamic programming algorithms for solving MDPs (Markov Decision Process), Policy Evaluation, Policy Iteration, Policy Improvement and Value Iteration.
4. Write a program to implement Dynamic programming algorithms for Policy Evaluation, Policy Iteration and Value Iteration.
5. Write a program to implement Grid-world by using Policy Iteration.
6. Write a program to perform Monte Carlo Prediction, Monte Carlo Off-Policy Control Importance Sampling.
7. Write a program to implement SARSA and Q-Learning in a grid-world or Frozen Lake environment based on Temporal Difference Learning.
8. Write a program to implement Cliff Walking problem using Q-learning.
9. Write a program to implement model-based planning with simulated experience for Dyna-Q Algorithm (Model-Based).
10. Write a program to implement a simple DQN agent using PyTorch or TensorFlow.
11. Write a program to implement Smart Traffic Light Controller using Reinforcement Learning.
12. Write a program to implement Cart Pole problem using a random policy and visualize how long it lasts.

Integrated Project (Mandatory) based upon the learnt concepts:

1. **Tic-Tac-Toe Game Agent using Q-Learning:** Develop an AI agent that learns optimal strategies for Tic-Tac-Toe through self-play and rewards-based learning using Q-values.
2. **Maze Solver using Q-Learning/SARSA:** To Train an agent to find the optimal path from the start point to the goal in a grid-based maze environment using model-free reinforcement learning algorithms.

Course Outcomes: After completion of this course, students will be able to:

1. Demonstrate various Components of Reinforcement Learning.
2. Make use of various exploration and exploitation strategies.
3. Apply Model based and Model Free Prediction techniques.
4. Make use of different value based Reinforcement Learning Algorithms.
5. Demonstrate various Policy based Reinforcement Learning Algorithms.